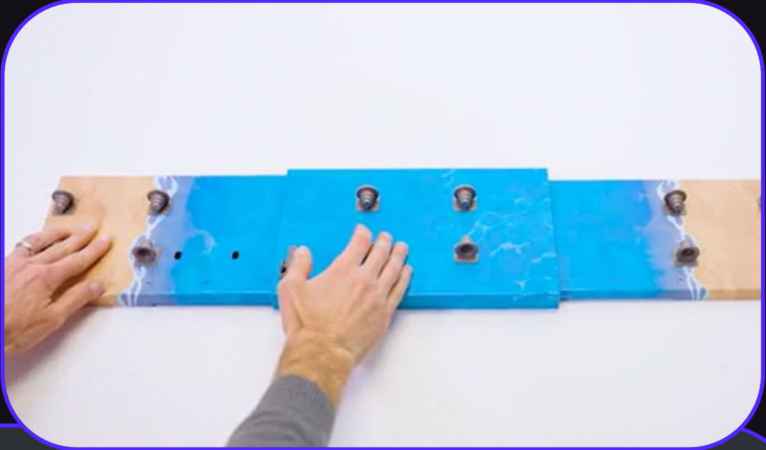
Bridge Mat

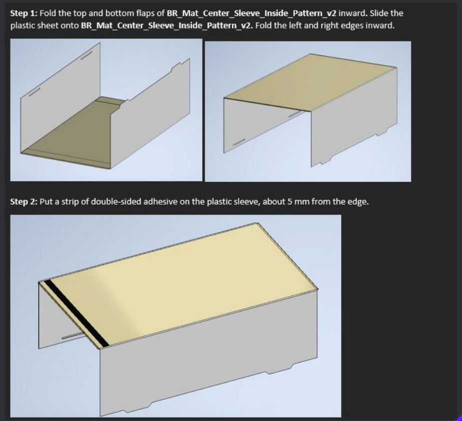
Updated Bridge or Bust’s game mat to:

- Reduce friction when paper wrap is added
- Increase stability with large bridges
- Simplify the design without losing function
- Make manufacturability easier for the producer



“I have not failed. I've just found 10,000 ways that won't work." - Thomas A. Edison

Learned how factory communication works and created a step by step guide to replicate my process. The guide is made to be efficient and repeatable for the factory.

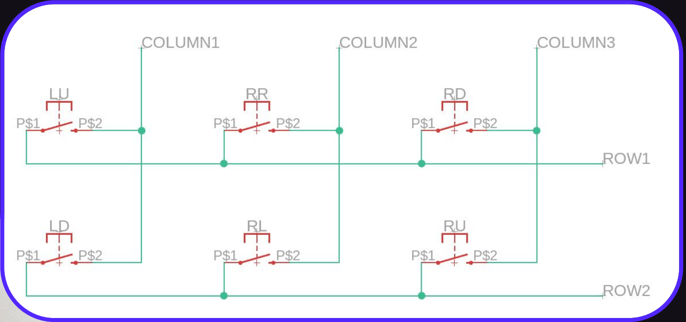


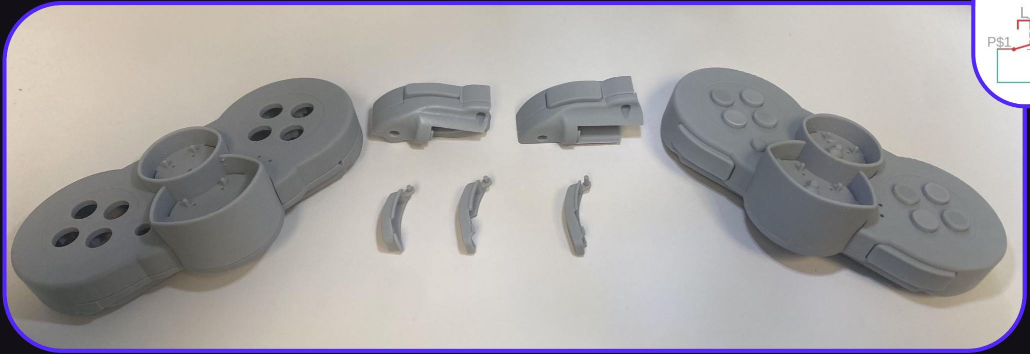
Step 1: Fold the top and bottom flaps of **BR_Mat_Center_Sleeve_Inside_Pattern_v2** inward. Slide the plastic sheet onto **BR_Mat_Center_Sleeve_Inside_Pattern_v2.** Fold the left and right edges inward.

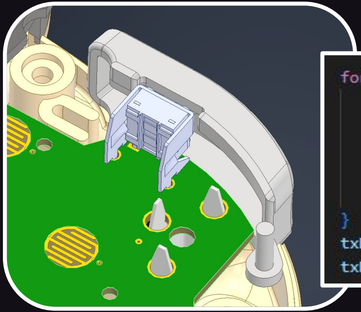
Step 2: Put a strip of double-sided adhesive on the plastic sleeve, about 5 mm from the edge.

Flying Car Remote

Integrated a bumper system to the remote allowing for more possible player inputs for the finalized version of the game.





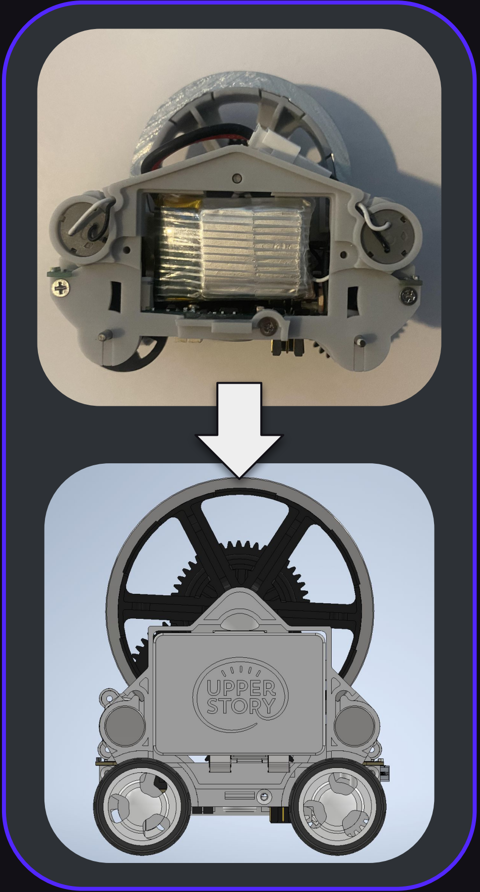


```
for(int r = 0; r < NUM_ROWS; r++){
   GPIO_write(index: rowPins[r], value: 0);
   for(int c = 0; c <NUM_COLS; c++){
    buttons |= !GPIO_read(index: columnPins[c]) << buttonValues[r][c];
   }
   GPIO_write(index: rowPins[r], value: 1);
}
txData[HDR_LEN] = buttons;
txData[HDR_LEN + 1] = buttons2;
```

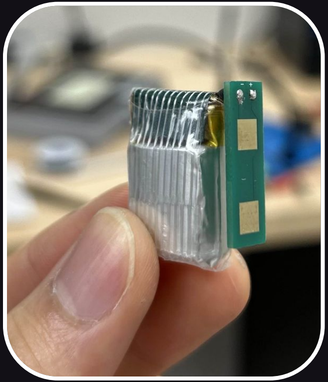
Learned how to fully implement a project into an existing system by:

- Prototyping the mechanical solution and iterating
- Develop a working schematic to simulate the system
- Update and find new electrical components for the PCB
- Execute the changes in the code

Removable Battery Solution

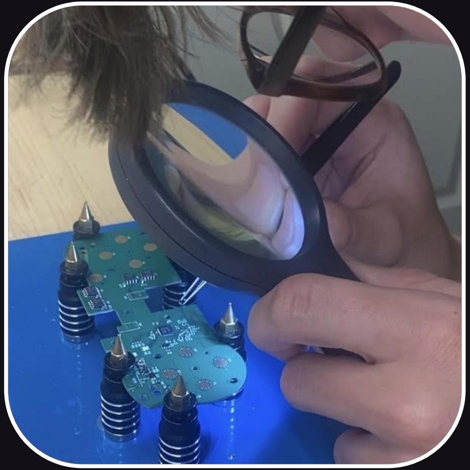


Designed a solution to enable non-stop play for the user by replacing batteries. The enclosed battery uses a custom PCB to connect to the inside of the car and battery charger.



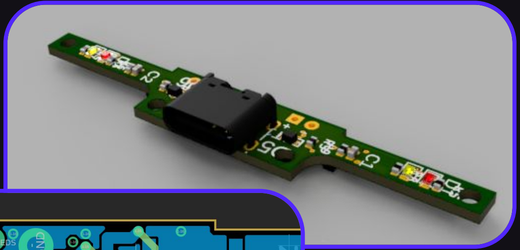
Learned how to work alongside someone on the same design allowing for changes to the battery and car frame to be cohesive and collaborative.

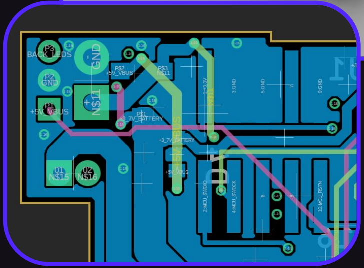
Printed Circuit Boards



Learned how to use Fusion 360s PCB editor to make custom PCB changes such as:

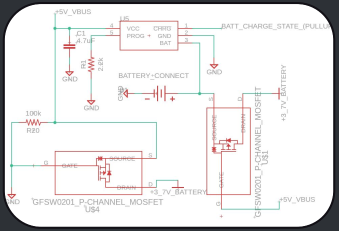
- Updating remote for bumper changes
- Making car battery health improved
- Designing battery connection board
- Prototyping charging station iteration



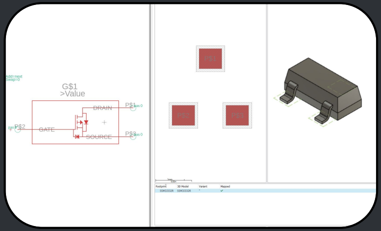


PCB Design Cycle

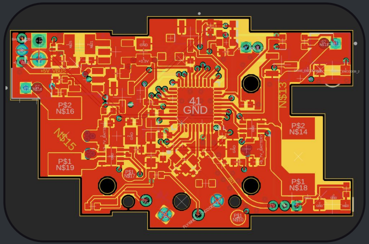
1: Build the schematic



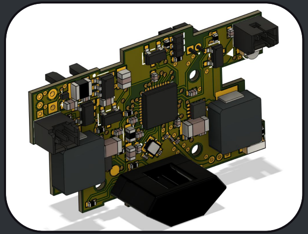
2: Design part footprint



3: Route PCB layers

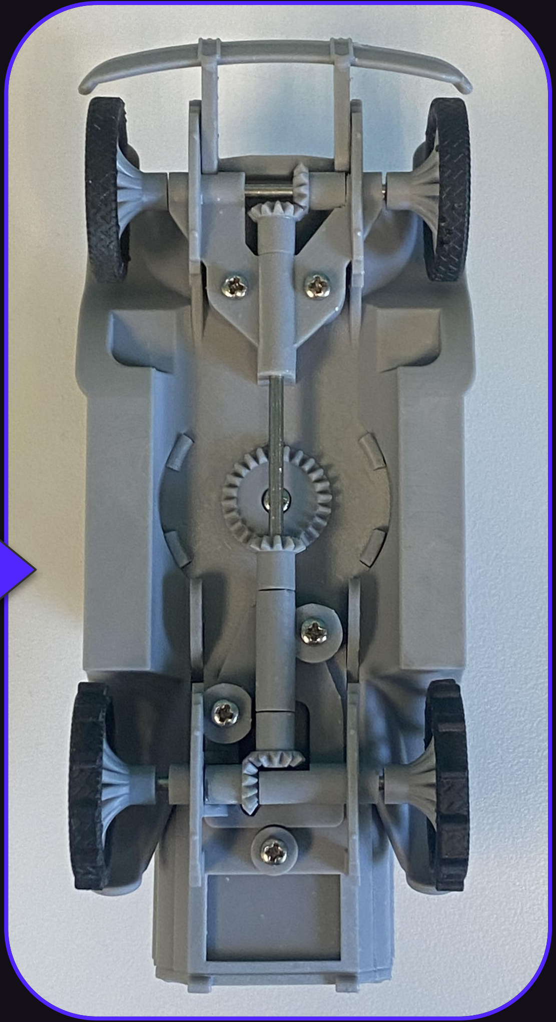


4: Produce circuit board



4WD Drivetrain

Prototyped a solution to integrate 4WD into the game's car. Created new and adjusted existing parts keeping in mind the constraints and ability to manufacture. Used resin printing to test and iterate through ideas. This change made the car able to climb steep roads and have the wheels slip less often.



Additionally, redesigned the wheels to the car to increase strength and reliability after the original design broke during testing.

Vacuum Molded Tray

Developed a vacuum tray to store game parts when packaging. Designed to optimize the most efficient way to store every part while minimizing material use and total volume. Tested and adjusted designs based on prototypes made from FDM 3D printing.